Arvapalli Venkata Sai Akhil

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B.Tech CSE (AI/ML) Student | Full-Stack AI Engineer

Education

Bachelor of Technology (B.Tech) in Computer Science Engineering (AI/ML) Sept 2023 – June 2027
CMR College of Engineering and Technology, Hyderabad CGPA: 8.27/10

Experience

Full-Stack AI Intern Jan 2026 – July 2026
DarexAI (Startup) Remote *(Completion Certificate)*

- Engineered and deployed production **AI Voice Calling Agents** for a Canada-based restaurant client using **Retell AI**, building a complete **restaurant automation system** from scratch – handling reservations, takeaway orders, and customer interactions via voice, backed by a database for order and session tracking, integrated with **n8n** workflows and **Cloudflare** for end-to-end business automation.
- Architected modular **Agentic AI** workflows and enterprise-level **AI Automations** using **n8n**, connecting disparate APIs to streamline business operations and data synchronization.
- Developed and fine-tuned specialized **RAG pipelines** to improve AI knowledge retrieval, response accuracy, and production performance for domain-specific use cases.

Full-Stack Developer – Throttle Theory 2026
Freelance Hyderabad *(GitHub)* | *(Live Site)*

- Designed and built a cinematic, scroll-driven marketing website for **Throttle Theory**, a premium automotive garage brand, using **Vite, React 18, TypeScript, Three.js (React Three Fiber), and GSAP ScrollTrigger** for immersive 3D and motion-based storytelling.
- Deployed and maintained the production site on **Vercel**, handling end-to-end delivery from architecture to live hosting.

Projects

PhysicsLens GitHub | Live Site

- Developed an AI-powered physics problem parser using a fine-tuned open-source LLM (Ollama, Llama/Qwen) to convert natural-language physics questions into structured representations for automated analysis.
- Designed a hybrid AI + deterministic pipeline combining LLM-based parsing with a rule-based Newtonian physics engine to solve mechanics problems and generate labeled SVG free-body diagrams for inclined planes, friction, Atwood pulleys, and projectile motion.
- Built a modular backend using Python, FastAPI, and SymPy with schema-based validation via Pydantic, exposing REST APIs for JSON/SVG generation, alongside a Node.js frontend for interactive problem submission and visualization.

Silent Link GitHub

- Developed an offline sign language translation system using TensorFlow/PyTorch CNNs for real-time bidirectional communication (sign-to-text, text-to-animation), achieving <100 ms latency via on-device inference.

GIS-Based Flood Risk Assessment GitHub

- Built an AI-driven geospatial platform integrating terrain, soil, and satellite data for flood prediction and disaster response planning.
- Designed scalable ETL pipelines using Google Earth Engine for real-time geospatial data ingestion, and built interactive Power BI dashboards to visualize flood risk zones and severity levels, enabling evacuation route planning and disaster response decision support.

Achievements

- **Second Prize – Hackathon (CONVERGENCE 2K25)** Feb 2025 (Credential)
  Secured second place at VNRVJIET's National Technical Symposium for developing **Silent Link**, an offline DL-based assistive communication system with gesture recognition.
- **Top 5,000 Teams – Build With India Hackathon** Google | 2025 (Credential)
  Ranked among top 5,000 out of 25,000 teams at Google Office Finale for innovation and problem-solving.
- Strengthened **DSA** proficiency via competitive programming on LeetCode and CodeChef, focusing on optimal solutions and edge-case handling.

Certifications

- Gen AI Leader — Google (Credential)
- Google Cloud Certified Associate Cloud Engineer — Google (Credential)
- Programming, Data Structures, and Algorithms Using Python — NPTEL, Jan.–Mar. 2024 (Credential)
- IBM Certifications — Cognitive Class (Credentials)
- Cisco Certifications (Networking) — Cisco Networking Academy (Credentials)
- Aerial Drone Technology Workshop (ADTW) — Mahindra University × TIHAN IITH

Technical Skills

- **Programming Languages:** Python, Java, Kotlin, C, C++, JavaScript, SQL, R
- **Mobile Development:** Flutter, Android (Kotlin), React Native
- **Backend Engineering:** Flask, FastAPI, REST APIs, Firebase, Redis, PostgreSQL, Supabase
- **AI/ML:** TensorFlow, PyTorch, Deep Learning, ANN Classification, Model Training & Inference
- **GenAI:** LLMs, RAG, Fine-tuning, Prompt Engineering, LangChain, LangGraph, Agentic AI (n8n, Retell AI), AI Coding Agents (Gemini, Claude)
- **Deep Learning Models:** ResNet, DenseNet
- **Data & Analytics:** Power BI, Data Visualization, Data Cleaning, ETL Pipelines
- **Cloud & Platforms:** Google Earth Engine, Cloudflare
- **Tools:** Git/GitHub, Android Studio, VS Code, Jupyter Notebook, Streamlit, Microsoft Power Apps

Coursework

Object Oriented Programming, Data Structures and Algorithms, DBMS, Web Development, Machine Learning, Artificial Intelligence, Nature Inspired Computing, Software Engineering, Operating Systems, Natural Language Processing, Data Analytics, Knowledge Representation and Reasoning, Automata Theory, Compiler Design, Image Processing, Computer Vision & Robotics